FIN-07-UNIT-ECONOMICS

FIN-07: Unit Economics — Project AEGIS / Carrington Storm Motors

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Executive Summary: The Unit Economics Thesis

Unit economics is where the engineering meets the financial model. A company can have a compelling total addressable market and a visionary CEO, but if it loses money on every unit it sells, it dies. CSM's unit economics demonstrate positive contribution margin on every product line by Year 1 and rapidly expanding unit profitability as manufacturing volumes scale. By Year 5, every major product line generates 40-60%+ gross margins at the unit level.

Williams Heuristic: If you sell something for $100 and it costs you $101 to make, you can't make it up in volume. If it costs you $70 to make, you have a business. If it costs you $40 to make, you have a great business. CSM's products cost between $40 and $82 to make, selling for $100. The range depends on the product and the year. The trajectory on every product is toward the $40 end. That's the unit economics thesis in one sentence.

Product Line 1: Aegis-C Composite Shielding (ACS)

Unit Definition: One transformer shield kit, custom-fit to transformer MVA rating class, including composite panels, grounding hardware, installation guide, and certification documentation. Average unit represents a medium-voltage (10-100 MVA) transformer application.

TABLE 7.1: ACS Unit Economics at Scale

Metric	100 Units (Y1)	1,000 Units (Y2/Y3)	10,000 Units (Y4/Y5)	100,000 Units (Y7+)
ASP (Average Selling Price)	$25,000	$24,250	$28,800	$30,000
Raw Materials	$10,000	$7,500	$6,000	$5,100
Direct Labor	$6,250	$4,500	$2,160	$1,080
Manufacturing Overhead	$3,750	$2,750	$1,920	$960
Packaging / Shipping / Install Support	$1,250	$1,000	$960	$720
Warranty Reserve (2%)	$500	$485	$576	$600
Total COGS per Unit	**$21,750**	**$16,235**	**$11,616**	**$8,460**
Gross Profit per Unit	**$3,250**	**$8,015**	**$17,184**	**$21,540**
Gross Margin %	**13.0%**	**33.0%**	**59.7%**	**71.8%**

Learning Curve Effect: 15% cost reduction per doubling of cumulative volume. The cost progression from 100 to 100,000 units (10 doublings) yields theoretical cost reduction of (0.85^10) = 19.7% of original cost. In practice, materials hit a floor price (raw polymer + dielectric filler costs), so the modeled reduction is shallower at high volumes.

TABLE 7.2: ACS Breakeven Analysis

Parameter	Value
Fixed Costs (per year, allocated to ACS line)	See below by volume
Breakeven Units (at Y1 unit economics)	Not applicable — variable cost below ASP from unit 1
Breakeven Revenue	Any positive revenue generates positive contribution margin
Time to Product-Level Profitability	Y1 (positive gross margin from first unit)
Time to Full-Cost Profitability (incl. allocated OpEx)	Y3 (~1,400 units × $12,600 GP = $17.6M vs. ~$8M allocated ACS OpEx)

El Segundo Calm: At 100 units, the unit margin is thin ($3,250 per kit) but positive. At 10,000 units, it's $17,184 per kit — more than 5x improvement. This is the magic of manufacturing learning curves combined with pricing power. The price doesn't fall as fast as the cost does. The spread widens. This is why we build factories.

Product Line 2: Aegis Home (AHM)

Unit Definition: One residential protection kit (main panel + 2 downstream circuits), packaged for DIY installation with professional-install option.

TABLE 7.3: AHM Unit Economics at Scale

Metric	200 Units (Y1)	1,000 Units (Y2)	10,000 Units (Y3)	100,000 Units (Y4/Y5)	1,000,000 Units (Y7+)
ASP	$2,500	$2,500	$3,000	$3,450	$3,500
Raw Materials (composite, electronics, enclosure)	$750	$600	$480	$380	$320
Direct Labor (assembly, testing, packaging)	$400	$320	$250	$180	$100
Manufacturing Overhead	$300	$240	$200	$150	$90
Packaging / Fulfillment	$200	$170	$150	$120	$90
Warranty Reserve (2%)	$50	$50	$60	$69	$70
Total COGS per Unit	**$1,700**	**$1,380**	**$1,140**	**$899**	**$670**
Gross Profit per Unit	**$800**	**$1,120**	**$1,860**	**$2,551**	**$2,830**
Gross Margin %	**32.0%**	**44.8%**	**62.0%**	**73.9%**	**80.9%**

Learning Curve: 20% cost reduction per doubling (consumer electronics-like learning, simpler manufacturing than ACS).

TABLE 7.4: AHM Channel Economics

Channel	% of Units (Y5)	ASP	COGS	GP per Unit	CAC	LTV (5yr)	LTV:CAC
DTC (csmshield.com)	35%	$3,590	$899	$2,691	$180	$3,200	17.8x
Insurance-Bundled	45%	$3,590	$899	$2,691	$80*	$3,500	43.8x
Utility Partnership	20%	$3,050	$899	$2,151	$120*	$2,800	23.3x

*CAC is lower in insurance and utility channels because the partner bears customer acquisition cost; CSM's cost is channel management and partner integration.

Williams Heuristic: In the DTC channel, we spend $180 to acquire a customer who generates $2,691 in gross profit on the first purchase — a 15x payback on acquisition cost within one transaction. In the insurance channel, the LTV:CAC is 44x because the insurance carrier acquires the customer, and we just fulfill. Any business where your best channel has a 44:1 LTV:CAC ratio is a business where you should invest every available dollar into that channel. The constraint is not CAC efficiency; it's the speed at which insurance carriers can onboard EMP rider products.

Product Line 3: Charlemagne-Class Fleet (CCF)

Unit Definition: One complete Charlemagne-class vessel, fully outfitted, delivered and certified.

TABLE 7.5: CCF Unit Economics — Per-Vessel

Metric	Unit #1 (Y1)	Unit #5 (Y2)	Unit #20 (Y3/Y4)	Unit #50 (Y5)
Contract Value (ASP)	$5,000,000	$6,000,000	$6,600,000	$7,200,000
Hull & Propulsion (subcontracted)	$1,800,000	$2,100,000	$2,200,000	$2,400,000
EMP Restoration Equipment	$1,500,000	$1,700,000	$1,650,000	$1,600,000
C4ISR Systems	$500,000	$600,000	$580,000	$550,000
Integration Labor & Engineering	$500,000	$520,000	$470,000	$420,000
Sea Trials, Certification, Delivery	$300,000	$300,000	$270,000	$250,000
Warranty Reserve (3%)	$150,000	$180,000	$198,000	$216,000
Total COGS	**$4,750,000**	**$5,400,000**	**$5,368,000**	**$5,436,000**
Gross Profit per Vessel	**$250,000**	**$600,000**	**$1,232,000**	**$1,764,000**
Gross Margin %	**5.0%**	**10.0%**	**18.7%**	**24.5%**

Build Time per Vessel: - Unit #1 (prototype): 18 months → $250K GP - Unit #5: 14 months → $600K GP - Unit #20: 12 months → $1.23M GP - Unit #50: 10 months → $1.76M GP

Learning Curve: 10% cost reduction per doubling (naval construction is inherently less scalable than composites manufacturing).

Product Line 4: Phantom MK-1 Robotics (PMR)

Unit Definition: One complete Phantom MK-1 system = 1 ground unit + 1 aerial unit + control station + 1-year software subscription.

TABLE 7.6: PMR Unit Economics at Scale

Metric	20 Units (Y1)	100 Units (Y2)	1,000 Units (Y3/Y4)	10,000 Units (Y7+)
ASP (Hardware + 1yr Software)	$50,000	$52,600	$58,000	$65,000
Hardware Components	$17,500	$15,500	$13,000	$10,000
Assembly & Testing Labor	$7,500	$6,500	$5,000	$3,500
Manufacturing Overhead	$4,000	$3,500	$2,800	$2,000
Software (capitalized per unit)	$2,500	$2,000	$1,500	$800
Packaging, Shipping, Training	$1,500	$1,500	$1,300	$1,000
Warranty Reserve (2%)	$1,000	$1,052	$1,160	$1,300
Total COGS	**$34,000**	**$30,052**	**$24,760**	**$18,600**
Gross Profit per Unit	**$16,000**	**$22,548**	**$33,240**	**$46,400**
Gross Margin %	**32.0%**	**42.9%**	**57.3%**	**71.4%**

TABLE 7.7: PMR Recurring Revenue Per Unit (Software)

Metric	Y3	Y4	Y5
Installed Base (cumulative systems)	475	1,395	3,245
ARPU (annual recurring per unit)	$5,556	$7,000	10, 811$\|\|SoftwareRevenue$(M)

Accountant Heuristic: The hardware gets us in the door. The software keeps us there. By Year 5, software revenue is 16.7% of total PMR revenue but 100% of PMR recurring revenue. The blended hardware+software margin improves because software COGS (server costs, maintenance) are minimal — $800/unit/year vs. $10,811 ARPU, a 92.6% incremental margin. This is the “razor and blades” model, defense-edition: the robot is the razor; the grid health analytics platform is the blades.

Product Line 5: Infrastructure Hardening (IFH)

Unit Definition: Average project = single substation retrofit. Project mix includes substations (50%), bridges (20%), data centers (15%), government facilities (15%).

TABLE 7.8: IFH Unit Economics — Per Average Project

Metric	Y1 ($500K)	Y2 ($500K)	Y3 ($600K)	Y4 ($650K)	Y5 ($710K)
Project Revenue (avg.)	$500,000	$500,000	$595,000	$647,000	$710,000
Materials	$175,000	$162,500	$178,500	$181,160	$184,600
Direct Installation Labor	$125,000	$118,750	$125,000	$123,500	$127,800
Engineering & PM	$50,000	$43,750	$47,600	$51,760	$56,800
Equipment & Tooling	$25,000	$25,000	$23,800	$25,900	$21,300
Permits, Inspections, Compliance	$15,000	$15,000	$17,850	$19,410	$21,300
Warranty Reserve (2%)	$10,000	$10,000	$11,900	$12,940	$14,200
Total COGS	**$400,000**	**$375,000**	**$404,650**	**$414,670**	**$426,000**
Gross Profit per Project	**$100,000**	**$125,000**	**$190,350**	**$232,330**	**$284,000**
Gross Margin %	**20.0%**	**25.0%**	**32.0%**	**35.9%**	**40.0%**

Customer Acquisition Cost (CAC) by Channel

TABLE 7.9: CAC Analysis

Channel	Y1 CAC	Y3 CAC	Y5 CAC	% of Revenue (Y5)
Government BD (per contract)	$85,000	$55,000	$45,000	38%
Utility Direct Sales (per customer)	$45,000	$30,000	$22,000	18%
Insurance Channel (per carrier)	$350,000	$200,000	$120,000	7%*
DTC (per customer)	$250	$180	$150	12%
Certified Installer Network (per installer)	$15,000	$8,000	$5,000	—**
Licensing/Partner (per licensee)	$50,000	$30,000	$20,000	—**

*Insurance channel revenue as % of total (IPP line only, not AHM through insurance channel). **Not a direct revenue line; supports ACS/IFH/AHM sales.

Lifetime Value (LTV) and LTV:CAC Ratios

TABLE 7.10: LTV by Customer Segment

Customer Segment	Avg. Annual	Avg. Relationship	Gross Margin	LTV	CAC	LTV:CAC	Payback (months)

	Revenue	(years)					
Federal Government (DoD, DOE, DHS)	$8.5M	8	25%	$17.0M	$350K	48.6x	5
State/Local Government	$1.2M	5	28%	$1.7M	$85K	19.8x	8
Utility (>1M customers)	$3.5M	7	40%	$9.8M	$120K	81.7x	4
Utility (<1M customers)	$0.4M	5	42%	$0.8M	$45K	18.7x	7
Insurance Carrier Partner	$0.6M	10	83%	$5.0M	$200K	24.9x	4
Aegis Home DTC Consumer	$3,590	5	74%	$2,830*	$180	15.7x**	1**
Aegis Home (via Insurance)	$3,590	8	74%	$2,830*	—	∞***	0

LTV based on initial hardware purchase only; does not include future accessory/upgrade purchases. ***Payback is immediate (within first transaction) because one-time hardware purchase exceeds CAC.*** Insurance channel has effectively $0 incremental CAC to CSM; carrier bears acquisition cost.

El Segundo Calm: The worst LTV:CAC ratio in our business is 15.7x (DTC consumer). Most SaaS companies would kill for a 15.7x LTV:CAC. The best is “infinite” — insurance carriers pay us to acquire their own customers for them. Any business where your worst channel is 15x and your best is free has a customer acquisition problem that is actually a “how do we scale this faster” problem, not a “does this work” problem.

Contribution Margin Analysis

TABLE 7.11: Contribution Margin by Product Line (Y5)

Product Line	Revenue per Unit	Variable COGS per Unit	Contribution per Unit	Contribution Margin %
ACS (composite shielding)	$30,000	$9,600*	$20,400	68.0%
AHM (Aegis Home)	$3,590	$720	$2,870	79.9%
CCF (Charlemagne vessel)	$7,200,000	$4,800,000*	$2,400,000	33.3%
PMR (Phantom system)	$64,900	$16,000	$48,900	75.3%
IFH (avg. project)	$710,000	$400,000	$310,000	43.7%
IPP (insurance)	—	—	—	~90%**
LIR (licensing)	—	—	—	~95%**

*Variable COGS only (materials + direct labor + shipping + warranty); excludes allocated fixed manufacturing overhead. **IPP and LIR have negligible variable COGS; essentially pure contribution margin businesses.

Williams Heuristic: Contribution margin is what’s left after paying for the stuff that goes into the product. Fixed costs (the factory, the engineers, the 18-agent command) come out of contribution margin. If contribution margin exceeds fixed costs, you have a business. If it doesn’t, you have a hobby. In Year 5, total contribution margin is ~$630M against $270M in OpEx (mostly fixed). That’s a 2.3x coverage ratio. The business works.

Payback Periods

TABLE 7.12: Capital Payback on Key Investments

Investment	Capital Outlay	Annual Cash Flow Generated	Payback Period
Automated Composite Layup Line	$2.5M	$1.7M (100 units × $17K GP)	~1.5 years
AHM Assembly Line	$1.5M	$2.8M (3,500 units × $800 incremental GP vs. contract mfg)	~6 months
Phantom Assembly Line	$1.0M	$1.5M (100 units × $15K GP at scale)	~8 months
International Office (one region)	$1.5M/year	$12M incremental revenue within 24 months	~18 months
Insurance Carrier Integration	$0.5M	$0.5-2M/year in premium revenue share	3-12 months

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Manufacturing Cost Reduction Trajectory

TABLE 7.13: Learning Curve Summary by Product

Product	Learning Rate	Cumulative Volume Y1-Y5	Cost Reduction Achieved	Floor Cost (Y7+)
ACS	15% per doubling	16,320 units	53% from Y1	~$7,500
AHM	20% per doubling	117,900 units	67% from Y1	~$600
CCF	10% per doubling	84 vessels	31% from Y1	~$4,500,000
PMR	18% per doubling	3,245 systems	58% from Y1	~$15,000
IFH	10% per doubling	302 projects	28% from Y1	~$350,000

Accountant Heuristic: Learning curves are not magic. They're the result of: (a) process improvement — figuring out better ways to do things each time; (b) tooling investment — building jigs, fixtures, and automation that reduce labor; (c) supply chain optimization — qualifying lower-cost suppliers and negotiating volume discounts; and (d) design for manufacturability — redesigning products to be easier/cheaper to make without sacrificing performance. All four of these require deliberate investment. The learning curve doesn't just "happen" — you have to fund it. Our model funds it.

End of FIN-07: Unit Economics — Carrington Storm Motors, Project AEGIS Next: FIN-08: Funding Strategy